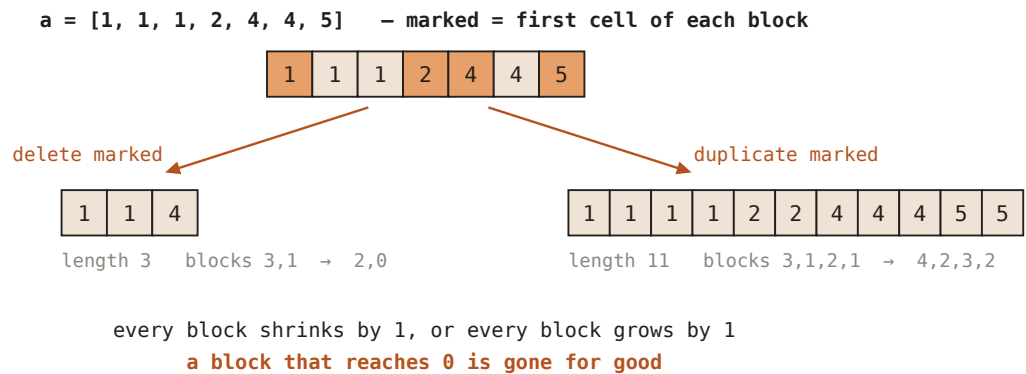


CF 2242C – Unstable Elements

how the solution works

THE OPERATION

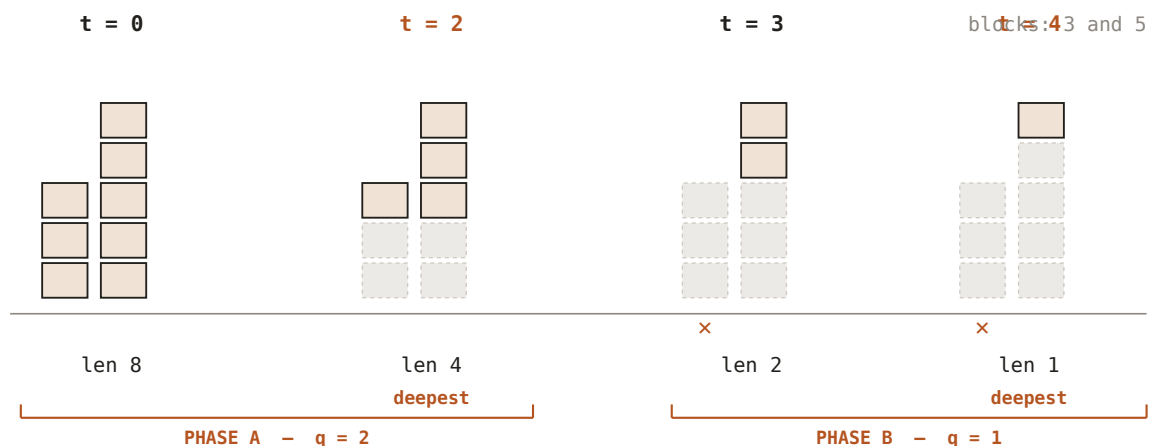
The array is sorted, so equal values sit together in blocks. One operation marks the first cell of every block and then either deletes all marks or duplicates them. Empty is forbidden. Count distinct reachable arrays of length k .



So the values never matter and the block boundaries never move. The whole array is just its list of block lengths.

ONE NUMBER: DEPTH

Delete subtracts 1 from every block, duplicate adds 1 to every block. Both act on all blocks at once, so track a single counter t = net deletions so far. A block of original length L currently holds $L - t$ cells.



t = how many deletions deep we are; shaded cells are eaten. The short block dies at $t = 3$, which splits the timeline into two phases.

DEATH IS ONE-WAY

When a block hits 0 it disappears, and duplicating cannot bring it back. **That asymmetry is the entire problem.** Blocks die shortest-first, so the set of dead blocks is always a prefix of the sorted lengths, and t can never again exceed the depth that killed them. Each dead set is a **phase**: inside it you slide t freely downward (duplicate, as far as you like) but only upward to the depth where the next block dies.

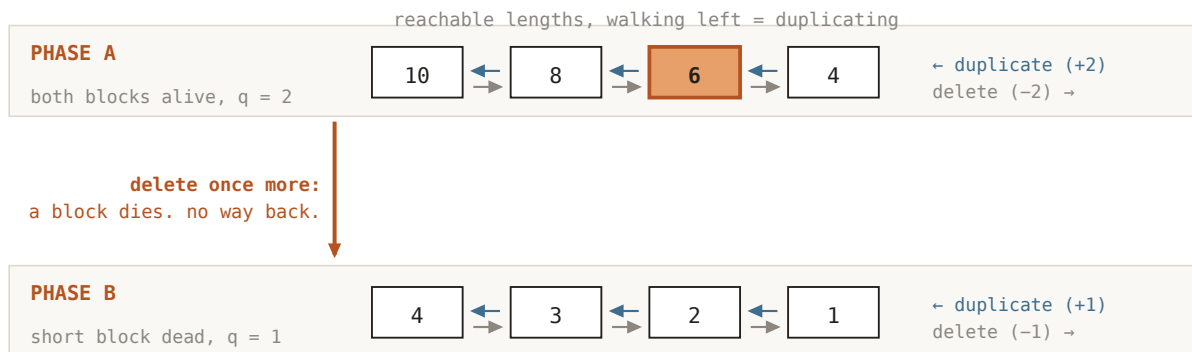
WHAT EACH PHASE CAN REACH

Fix a phase. Let q = alive blocks, d = total cells of the dead ones, len = shortest surviving original length. The deepest allowed depth is $t = len - 1$, giving the shortest array of the phase:

$$cl = n - d - (len - 1) \cdot q$$

From there each duplicate adds exactly q cells, forever. So the phase reaches lengths

$cl, cl + q, cl + 2q, cl + 3q, \dots$



$k = 6$ is hit in phase A ($6 = 4 + 1 \cdot 2$) and in phase B ($6 = 1 + 5 \cdot 1$) → answer 2

phase A gives [1,1,2,2,2,2] phase B gives [2,2,2,2,2,2]

the two phases hold different values, so they can never collide.

COUNTING

Two phases always differ in which values survive, so they never produce the same array. And inside one phase a length pins down t exactly. Therefore each phase contributes **at most one** array of length k , and it contributes one exactly when

$$cl \leq k \quad \text{and} \quad (k - cl) \bmod q = 0$$

Phases correspond one-to-one with the *distinct* block lengths, so:

- > collect block lengths, sort them ascending
- > sweep the distinct values; at each one q = blocks not yet dead, d = cells already dead
- > test the two conditions, count the hits

Answer is the number of hits. $O(n \log n)$, and it fits in 32 bits because there are at most $O(\sqrt{n})$ distinct block lengths.